

Guarded Explanations: Conformal-Style Filtering for Distribution-Aware Rule Conditions

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Abstract

In high-stakes decision support, post-hoc explanation methods are often used to estimate feature influence by perturbing inputs. However, a fundamental limitation of this approach is that candidate perturbations frequently fall outside the support of the calibration data, yielding unsupported and potentially misleading explanations. We introduce *guarded explanations*, an extension of the Calibrated Explanations framework. Guarded explanations apply a conformal-anomaly-detection-inspired filter to candidate perturbations, retaining a representative candidate only when its empirical guard score meets a user-chosen threshold ε . To facilitate this, we replace the standard binary discretiser with a multi-bin discretiser, generating interval-valued conditions alongside one-sided thresholds. Emitted perturbations are then scored using the standard Calibrated Explanations uncertainty-aware backend. In a synthetic constraint setting, we demonstrate that guarded factual explanations reduce the fraction of emitted rules whose representative values violate a known domain constraint from approximately 2.5% to near 0.0% at $\varepsilon = 0.2$. Crucially, we isolate this effect from discretisation granularity alone by comparing against an unpruned multi-bin (no-guard) baseline. We frame guarded explanations as a representative-level plausibility filter, where the guard demonstrably reduces calibration-unsupported representative candidates in emitted rules. The guard score is a conformal-style empirical rank, not a standard conformal p-value, and ε has no finite-sample error-rate guarantee for constructed perturbations.

Keywords: Conformal prediction, calibrated explanations, conformal anomaly detection, post-hoc explanations, distribution awareness, perturbation filtering

1. Introduction

In decision-support applications ranging from medical diagnosis to credit scoring, predictive models must be accompanied by explanations that practitioners can trust. Post-hoc explanation methods, such as LIME (Ribeiro et al., 2016), SHAP (Lundberg and Lee, 2017), and counterfactual generators (Wachter et al., 2018), answer why a model made a specific prediction by perturbing input features and observing the local effect on the model’s output.

A fundamental weakness of this approach, however, is distributional ignorance. Methods based on perturbations frequently explore the feature space without regard for the underlying data manifold. When a perturbation moves an instance into a low-density region, the model’s response is an extrapolation rather than an interpolation. For example, presenting a rule that alters an applicant’s years of professional experience to 40 while retaining an age of 20 may be mathematically valid for the model, but it is a semantically unsupported query. Generating explanations from uncalibrated, out-of-distribution regions introduces unseen risks into the decision-support process.

The relevant question is not merely how to generate plausible counterfactuals, but how to filter post-hoc explanation probes against observed calibration data, so that emitted rules are grounded in the support of the training distribution at the representative level. Existing approaches such as Diverse Counterfactual Explanations (DiCE) (Mothilal et al., 2020) or actionable recourse frameworks (Ustun et al., 2019) address distributional plausibility by requiring user-defined constraints or assumed causal structures, neither of which is readily available in general-purpose decision support.

We propose *guarded explanations*, an extension of the Calibrated Explanations framework (Löfström et al., 2024) that filters perturbation candidates against the calibration distribution before rules are assembled. Inspired by conformal anomaly detection (Laxhammar and Falkman, 2015), we introduce a model-agnostic guard that forwards only representative perturbations whose empirical rank score meets a user-chosen threshold ε to the prediction backend.

The contributions of this paper are strictly bounded to the representative level:

1. We introduce a conformal-style guard that filters individual perturbation candidates prior to rule assembly, alongside a multi-bin discretiser that expands the Calibrated Explanations rule language.
2. We empirically evaluate representative-level rule plausibility, showing that multi-bin discretisation and guard filtering together reduce constraint-violation rates to near zero. The guard provides the additional selective filtering needed to drive violations below the multi-bin no-guard baseline.
3. We evaluate guard-score discrimination, confirming that Fisher-combined candidate-level guard scores carry distributional signal under controlled shift, with discrimination strength depending on shift magnitude and dimensionality.
4. We evaluate on real-world benchmark datasets to verify that guards behave as expected when applied to benchmark datasets.

2. Related Work

Distribution awareness in explanations has been approached from several directions. Feasibility constraints in DiCE (Mothilal et al., 2020) and density-path search in FACE (Poyiadzi et al., 2020) are counterfactual-specific methods that require explicit constraint specification or density estimation. Actionable recourse frameworks (Ustun et al., 2019; Karimi

et al., 2021) rely on causal graphs. Laugel et al. (2019) demonstrate that post-hoc explanations can produce rules grounded in low-density regions, but propose diagnosis rather than prevention. Slack et al. (2021) propose reliability indicators but do not filter perturbations.

Similar concerns arise in feature-attribution methods under feature dependence, where marginal perturbations may create unrealistic feature combinations; conditional and data-manifold variants of Shapley explanations address related support-awareness issues (Aas et al., 2021).

Conformal prediction has been used in explainability contexts. Johansson et al. (2018) combine conformal prediction with rule extraction. Our work is distinct in that we use conformal anomaly detection as a *filter* on candidate perturbations within an existing explanation framework, rather than as a prediction or explanation method itself.

The broader conformal anomaly detection literature (Laxhammar and Falkman, 2015) provides the theoretical foundation for our guard. Our contribution is the integration of this detection step into the perturbation-based explanation pipeline of Calibrated Explanations (Löfström et al., 2024).

3. Background

3.1. Calibrated Explanations

Calibrated Explanations is a framework for generating post-hoc explanations of black-box model predictions with calibrated uncertainty quantification (Löfström et al., 2024). The framework supports binary and multiclass classification (via Venn-Abers predictors) and regression (via conformal predictive systems). It also supports probabilistic explanations for regression, combining both conformal predictive systems and Venn-Abers predictors. These probabilistic explanations explain the probability that the target is below a threshold (Löfström et al., 2025) or within an interval (Löfström et al., 2026).

Factual explanations answer “why did the model predict this?” by identifying which feature values, if changed, would most alter the prediction. For each feature, the instance value is perturbed to a representative value from a discretisation bin, and the calibrated prediction change (with uncertainty interval) is reported as the feature weight. In the standard Calibrated Explanations pipeline, a binary discretiser (`max_depth=1`) produces one-sided threshold conditions such as “age ≤ 30 ” or “age > 30 ”.

Alternative explanations answer “what would need to change?” by presenting candidate feature value changes that would shift the prediction, each with its own calibrated prediction and uncertainty interval.

A critical property of Calibrated Explanations is that prediction intervals are calibrated, meaning the uncertainty bounds reflect the true coverage probability under exchangeability. This distinguishes Calibrated Explanations from methods such as LIME or SHAP, which report point estimates without calibrated uncertainty.

3.2. Conformal-style anomaly score

Conformal anomaly detection (Laxhammar and Falkman, 2015) motivates the guard. Given a calibration set $D_{\text{cal}} = \{z_1, \dots, z_n\}$ and a non-conformity measure A , the implementation

evaluated here uses the unsmoothed empirical tail-rank score

$$p_{\text{guard}}(z) = \frac{|\{i : A(z_i) \geq A(z)\}|}{n}, \quad (1)$$

where $A(z_i)$ is the non-conformity score of calibration point z_i , computed using leave-one-out to avoid self-referencing, while $A(z)$ is computed against the full calibration set. A candidate is declared *conforming* if $p_{\text{guard}}(z) \geq \varepsilon$, where $\varepsilon \in (0, 1)$ is the guard threshold. Because the calibration set is used both as the KNN reference set and to form the leave-one-out score distribution, and because constructed perturbations are not exchangeable calibration observations, p_{guard} is not a standard conformal p-value. In particular, ε is a filtering parameter and does not provide a finite-sample bound on the probability of rejecting a constructed perturbation. We retain the conformal terminology only to identify the rank-based anomaly-detection construction that motivates the guard; conformal p-values should otherwise be interpreted as distributional evidence rather than ordinary probability estimates (Bates et al., 2023).

We use the distance to the k -th nearest neighbour as the non-conformity score:

$$A(z) = d(z, z_{(k)}), \quad (2)$$

where $z_{(k)}$ is the k -th nearest calibration point. The experiments use Euclidean distance with $k = 5$. Numeric features are min-max normalised before computing distances, and categorical features are one-hot encoded for the distance calculation. We refer to the combination of this KNN score, the empirical rank in (1), and thresholding at ε as the *KNN guard*.

3.3. Interpretation and limitations

It is useful to separate the guard score from the guarantees of the Calibrated Explanations backend. The guard itself has no finite-sample rejection guarantee. Guarded explanations reuse the unchanged Calibrated Explanations prediction backend for retained candidates, but guard-based selection does not create a new conditional or per-rule coverage guarantee. Any coverage statement for a backend prediction continues to require that backend’s own assumptions, including exchangeability of the queried point with its calibration population.

These statements do not imply causal or actionability guarantees and do not hold under distribution drift. A synthetic perturbation created during explanation is a *constructed* query, so exchangeability is not inherited from the data-generating process. The guard therefore acts solely as a heuristic filter for calibration-supported perturbation probes. It should not be interpreted as a certificate of semantic realism, causal feasibility, or whole-interval plausibility.

4. Contribution: Guarded Explanations

Given a trained model f , a calibration set $D_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$, a test instance $x \in \mathbb{R}^d$, where d denotes the number of input features, a guard threshold $\varepsilon \in (0, 1)$, and a mode $m \in \{\text{factual}, \text{alternative}\}$, the method uses the standard Calibrated Explanations prediction pipeline to produce rules from perturbations x' whose guard scores satisfy

$$p_{\text{guard}}(x' \mid D_{\text{cal}}) \geq \varepsilon, \quad (3)$$

Table 1: Rule condition formats produced by multi-bin discretisation.

Position	Format	Example
First bin ($l = -\infty$)	feature $\leq u$	age ≤ 30
Interior bin	$l < \text{feature} \leq u$	$30 < \text{age} \leq 50$
Last bin ($u = +\infty$)	feature $> l$	age > 50
Categorical	feature = ‘value’	colour = ‘red’

Algorithm 1 Guarded Explanation

Require: Model f , calibration set D_{cal} , instance x , threshold ε , mode m , neighbours k
Ensure: Set of guarded explanation rules $\mathcal{R}$

- 1: Build KNN guard on calibration features X_{cal} with k neighbours
- 2: Discretise each feature into bins via decision tree with `max_depth=3`
- 3: **for** each feature f_i and bin b_j **do**
- 4: Construct perturbation $x'_{i,j} = (x_1, \dots, x_{i-1}, r_j, x_{i+1}, \dots, x_d)$
- 5: Compute guard score $p_{i,j} = p_{\text{guard}}(x'_{i,j} \mid D_{\text{cal}})$ via (1)-(2)
- 6: **if** $p_{i,j} \geq \varepsilon$ **then**
- 7: Mark b_j as *conforming*
- 8: **else**
- 9: Mark b_j as *removed*; record in audit trail
- 10: **end if**
- 11: **end for**
- 12: **(Optional)** Merge adjacent conforming bins with re-testing
- 13: Batch-predict all conforming perturbations through the Calibrated Explanations backend to obtain $\hat{y}_{i,j}$ with intervals
- 14: **if** $m = \text{factual}$ **then**
- 15: Emit the factual bin for each feature (the bin containing x ’s value)
- 16: **else**
- 17: Emit all non-factual conforming bins as alternative rules
- 18: **end if**
- 19: **return** $\mathcal{R}$

The discretisers fit a univariate decision tree for each feature; `max_depth` is the tree’s maximum depth. Standard factual explanations use `max_depth=1`, producing at most two bins and a single threshold split per feature. Guarded explanations use `max_depth=3`, producing up to $2^3 = 8$ bins per feature. This extends the rule language as shown in Table 1.

For each feature f_i , the discretiser produces bins $\{b_1, \dots, b_{B_i}\}$ where each bin b_j is defined by interval bounds $[l_j, u_j)$. The representative value r_j for each bin is the median of the calibration samples falling within that bin.

Algorithm 1 summarises the guarded explanation procedure.

For each feature f_i and bin b_j , the perturbed instance

$$x'_{i,j} = (x_1, \dots, x_{i-1}, r_j, x_{i+1}, \dots, x_d) \quad (4)$$

changes only feature i while keeping all other features at their original values. Although one feature is changed at a time, the KNN score is computed on the complete perturbed vector $x'_{i,j}$ against multivariate calibration points. It can therefore flag an unsupported combination such as age 20 with 40 years of experience; it is not a univariate marginal-density test. The method does not, however, certify simultaneous changes to multiple features. The bin is retained if and only if $p_{i,j} \geq \varepsilon$.

All conforming perturbations are collected into a single batch and scored through the calibrated prediction pipeline, yielding for each:

- A point prediction $\hat{y}_{i,j}$
- A prediction interval $[\hat{y}_{i,j}^{\text{low}}, \hat{y}_{i,j}^{\text{high}}]$

Batching avoids redundant calibration computations.

Adjacent conforming bins can be merged into wider intervals. For each group of adjacent conforming bins, the algorithm (1) computes a merged representative from calibration values in the combined interval, (2) re-tests the merged representative via the guard, and (3) accepts the merged bin only if the re-test passes. In alternative mode, merging never crosses the factual bin, preserving the distinction between the current value and candidate changes.

In *factual mode*, only the factual bin (containing the instance’s observed value) is reported per feature. The feature weight is the calibrated prediction difference between the factual-bin representative and the original instance, with its uncertainty interval. In *alternative mode*, all non-factual conforming bins are reported as candidate changes, each with its own rule condition, prediction, and interval.

5. Evaluation Method

5.1. Representative-Level Rule Plausibility Under a Known Constraint

We generate a binary classification task in dimensionalities $d \in \{2, 10\}$ where all training, calibration, and test instances satisfy the domain constraint $x_2 \leq 2x_1 + 3$. Two models are trained, namely logistic regression and random forest. In the paper-facing configuration, we explain 500 sampled test instances per seed (with 12 seeds) with baseline Calibrated Explanations and guarded Calibrated Explanations at guard thresholds $\varepsilon \in \{0.01, 0.05, 0.1, 0.2\}$, with $k = 5$ neighbours, $n_{\text{train}} = 5,500$, and $n_{\text{cal}} = 3,200$. We evaluate factual mode (“why this prediction?”) explanations.

The *violation rate* is the fraction of emitted rules whose representative perturbation point violates the constraint. Lower is better. This evaluation is representative-level only. We score only the representative values from emitted records and do not claim whole-interval validity.

5.2. Guard-Score Discrimination Under Controlled Shift

The diagnostic uses Fisher-combined guard scores as an instance-level anomaly summary for evaluation purposes only. When the guard processes an instance, it assigns a per-candidate conformity score to each representative perturbation. Fisher combination aggregates these

candidate-level scores into a single per-instance statistic, $\hat{p}_{\text{Fisher}}$. This combined score is used here only as a ranking device for AUROC computation; it is not a conformal p-value, and it is not proposed as a deployed anomaly detector. The purpose is to test whether candidate-level guard conformity scores are systematically lower for distributionally shifted instances than for in-distribution instances.

For each of 10 seeds and dimensionality $d \in \{2, 5, 10, 20\}$, we train a classifier on $n_{\text{train}} = 5,000$ instances and build a guard on $n_{\text{cal}} = 3,000$ calibration instances. We sample 1,000 in-distribution (ID) test instances and 1,000 out-of-distribution (OOD) instances per shift level (mild: 1σ , moderate: 2σ , extreme: 5σ), with the shift vector rescaled by $1/\sqrt{d}$ to keep total shift magnitude comparable across dimensionalities. OOD instances are sampled independently from a Gaussian shifted by the shift vector, not by perturbing ID instances, to avoid inflated AUROC from ID/OOD feature correlation. Mean AUROC of $(1 - \hat{p}_{\text{Fisher}})$ is reported as a diagnostic anomaly score for separating ID from shifted samples, averaged over 10 seeds at $k = 5$ neighbours with normalised guard scores.

5.3. Evaluation with Benchmarking Datasets

In addition to the synthetic and controlled-shift evaluations, we assess the guard on a wide suite of publicly available benchmark datasets. Table 2 lists over thirty datasets spanning binary and multiclass classification as well as standard and probabilistic regression tasks, with sample counts ranging from 768 to 20 640 and feature dimensionalities between 2 and 40. Each dataset is randomly partitioned into training, calibration, and test sets, with 100 test instances, three calibration set sizes and remaining data as training set (at least 100 instances). Random forests are used as underlying models. The guard is built on the calibration set and explanations. We evaluate guard thresholds $\varepsilon \in \{0.05, 0.1, 0.2\}$ and aggregate results over 3 random seeds and multiple calibration sizes (100, 300, and 500).

For factual explanations we report three metrics: the mean number of emitted rules per instance for the standard Calibrated Explanations and for guarded explanations, the *retention rate* defined as the ratio of emitted to total (emitted + guard-removed) factual bins, and the fraction of test instances for which no guarded factual rules are returned. For alternative explanations we report the mean number of alternative rules per instance, retention, and the fraction of instances with no alternatives. Because multi-bin discretisation can produce multiple interval-valued conditions per numeric feature, alternative-mode results may contain more rules than the standard calibrated explanations, only allowing one-sided conditions. This evaluation is a behavioral check of threshold monotonicity and retention across tasks; without ground-truth plausibility labels, it is not evidence of real-data plausibility benefit.

6. Evaluation

A plausibility filter for perturbation-based explanations must satisfy two conditions to be practically useful. Emitted rules should respect the support of the calibration data, and the underlying guard scores should carry meaningful distributional signal. We evaluate both conditions directly.

The plausibility evaluation tests whether emitted guarded factual rules, evaluated at the representative level, better respect a known structural constraint than standard Calibrated

Table 2: Datasets used in the benchmark evaluation. Columns are repeated three times to fit many datasets in one table: DS = dataset, t = task code (b = binary, m = multiclass, r = regression), n = instances, f = features.

DS	t	n	f	DS	t	n	f	DS	t	n	f
diabetes	b	768	8	german	b	955	27	kc1	b	1192	21
pc4	b	1343	37	ttt	b	958	27	cars	m	1728	6
cmc	m	1473	9	cool	m	768	8	heat	m	768	8
image	m	2310	19	steel	m	1941	27	vehicle	m	846	18
vowel	m	990	11	wave	m	5000	40	wineR	m	1599	11
wineW	m	4898	11	yeast	m	1484	8	abalone	r	4177	8
anacalt	r	4052	7	bank8fh	r	8192	8	bank8fm	r	8192	8
bank8nh	r	8192	8	bank8nm	r	8192	8	comp	r	8192	12
concreate	r	1030	8	cooling	r	768	8	deltaA	r	7129	5
deltaE	r	9517	6	friedm	r	1200	5	heating	r	768	8
housing	r	20640	9	kin8fh	r	8192	8	kin8fm	r	8192	8
kin8nh	r	8192	8	kin8nm	r	8192	8	laser	r	993	4
mg	r	1385	6	mortgage	r	1048	15	plastic	r	1649	2
puma8fh	r	8192	8	puma8fm	r	8192	8	puma8nh	r	8192	8
puma8nm	r	8192	8	quakes	r	2178	3	stock	r	950	9
treasury	r	1048	15	wineRed	r	1599	11	wineWhite	r	4898	11
wizmir	r	1461	9								

Explanations rules, in a synthetic setting where the constraint is exactly known. Guard-score discrimination tests whether the Fisher-combined guard scores carry distributional signal under controlled shift. Together, these two experiments support the paper’s narrow central claim: guarded explanations act as a selective, representative-level plausibility filter in the evaluated settings. As a complement, the benchmark evaluation characterises retention and sparsification behavior across real-world datasets without claiming ground-truth plausibility.

6.1. Representative-Level Rule Plausibility Under a Known Constraint

Table 3 shows the factual-mode violation rates. Standard Calibrated Explanations has a baseline violation rate of approximately 2.5 % regardless of model. Guarded Calibrated Explanations reduces violations to near 0.0 % at $\varepsilon = 0.2$ and below 0.2 % at $\varepsilon = 0.1$. Even the least aggressive threshold ($\varepsilon = 0.01$) reduces the violation rate to below 0.4 %.

Table 4 shows the factual-mode rule count, which captures the cost of filtering. Standard Calibrated Explanations emits approximately 5.4 rules per instance. Guarded Calibrated Explanations emits fewer rules as the guard threshold increases, declining monotonically with ε , from roughly 5.5 at $\varepsilon = 0.01$ down to roughly 4.1 at $\varepsilon = 0.2$. The reduction in violation rate is substantially larger than the reduction in rule count, indicating that the guard preferentially removes implausible rules.

Guarded Calibrated Explanations uses multi-bin discretisation (`max_depth=3`), while standard Calibrated Explanations uses binary discretisation (`max_depth=1`). To separate the two contributions, we include a “multi-bin (no guard)” ablation that applies multi-bin discretisation without activating the guard filter; this configuration appears as the third column in Table 3. The ablation shows that multi-bin discretisation alone reduces the

Table 3: Factual-mode violation rate: fraction of emitted rules violating the domain constraint $x_2 \leq 2x_1 + 3$. Mean over seeds and dimensions. “Multi-bin (no guard)” isolates the effect of discretisation granularity alone.

ε	Model	Guarded	Multi-bin (no guard)	Standard
0.01	Logistic Reg.	0.004	0.005	0.025
0.01	Random Forest	0.003	0.005	0.024
0.05	Logistic Reg.	0.001	0.005	0.025
0.05	Random Forest	0.001	0.005	0.024
0.10	Logistic Reg.	0.001	0.005	0.025
0.10	Random Forest	0.001	0.005	0.024
0.20	Logistic Reg.	<0.001	0.005	0.025
0.20	Random Forest	<0.001	0.005	0.024

Table 4: Factual-mode rule count per instance, mean over seeds and dimensionalities.

ε	Model	Guarded rules	Standard rules
0.01	Logistic Reg.	5.43	5.35
0.01	Random Forest	5.65	5.49
0.05	Logistic Reg.	5.10	5.35
0.05	Random Forest	5.22	5.49
0.10	Logistic Reg.	4.77	5.35
0.10	Random Forest	4.82	5.49
0.20	Logistic Reg.	4.15	5.35
0.20	Random Forest	4.11	5.49

representative violation rate from approximately 2.5% to approximately 0.5%, as finer bins yield representatives closer to the observed instance, which are less likely to cross a constraint boundary. The guard then drives violations from the 0.5% multi-bin baseline to near zero at $\varepsilon = 0.2$. The improvement from standard to guarded Calibrated Explanations therefore has two components. Richer rule discretisation accounts for the large step from 2.5% to 0.5%, and guard-based candidate filtering accounts for the additional reduction to near zero. This shows that in the synthetic constraint setting, guarded explanations do improve representative-level rule plausibility, with the guard providing the final reduction beyond finer discretisation alone.

6.2. Guard-Score Discrimination Under Controlled Shift

The rule plausibility evaluation shows that the guard reduces representative constraint violations in one constrained synthetic task, but does not directly test whether guard scores carry general distributional signal. A guard that achieved low violation rates primarily through the discretisation geometry, rather than its rank score, would not necessarily respond to distributional distance. Guard-score discrimination evaluation therefore provides a targeted diagnostic that tests whether the guard scoring mechanism responds monotonically to controlled distribution shift in a setting where the correct answer is known by construction.

Table 5: Mean AUROC of Fisher-combined guard score ($1 - \hat{p}_{\text{Fisher}}$) used as a diagnostic anomaly score, by dimensionality and shift level ($k = 5$, `normalize_guard=True`, mean over 10 seeds).

Features (d)	Mild (1σ)	Moderate (2σ)	Extreme (5σ)
2	0.608	0.812	0.998
5	0.585	0.756	0.975
10	0.553	0.694	0.941
20	0.540	0.648	0.900

Table 5 shows mean AUROC by dimensionality and shift level. Extreme shifts (5σ) are strongly separable across all dimensionalities, with AUROC between 0.900-0.998, confirming that guard scores respond clearly to large distributional displacement. Moderate shifts (2σ) are useful at lower dimensionalities but degrade with increasing dimensionality, going from AUROC 0.812 at $d = 2$ to 0.648 at $d = 20$. Mild shifts (1σ) approach random ranking at higher dimensionalities, which is expected for distance-based KNN scoring since calibration neighbourhoods become sparser in high dimensions and distances less discriminative. These results support that the guard scores carry distributional information.

6.3. Evaluation with Benchmarking Datasets

Results from benchmarking datasets are divided into factual and alternatives and are shown in Tables 6 and 7. Regression results are from the same set of datasets, with the *Regr. plain* showing results from conformal prediction (using conformal predictive systems) and *Regr. p[...]* showing results from the probabilistic explanations using the 25th, 50th, and 75th target percentiles as thresholds.

Table 6: Guard retention benchmark for *factual* explanations across all tasks and regression modes (averaged over datasets, seeds, and calibration sizes). *Standard*: mean Calibrated Explanations rules per instance (no guard, at $\varepsilon = 0.10$); *Guarded*: mean guarded rules per instance; *Retention*: emitted / (emitted + guard-removed) over factual bins ($1 - \varepsilon$ is a descriptive comparison only, not a guaranteed floor); *Filtered*: fraction of instances with zero guarded rules.

ε		0.05	0.1	0.2	0.05	0.1	0.2	0.05	0.1	0.2
Task	Standard	<i>Guarded</i>			<i>Retention</i>			<i>Filtered</i>		
Binary	17.1	15.5	14.7	12.5	0.94	0.89	0.76	0.06	0.09	0.18
Multiclass	11.8	13.8	13.0	11.6	0.96	0.90	0.81	0.02	0.06	0.13
Regr. plain	7.6	7.2	6.8	5.9	0.95	0.90	0.79	0.02	0.05	0.12
Regr. p25	6.7	6.7	6.2	5.3	0.95	0.89	0.77	0.02	0.06	0.13
Regr. p50	6.7	6.8	6.3	5.4	0.95	0.89	0.78	0.02	0.06	0.13
Regr. p75	6.8	6.9	6.4	5.5	0.95	0.89	0.78	0.02	0.06	0.13

Table 6 summarises guard behavior across binary, multiclass and regression tasks for factual explanations. Since factual results only omit at most one rule per feature, the number of standard rules mirrors the average number of features per group of datasets.

Guarded explanations emit slightly fewer rules on average. However, as there is some filtering happening also for standard rules (e.g., not creating rules where there is no support), the guarded solution sometimes produces even slightly more rules than standard Calibrated Explanations. As the guard threshold increases, the number of factual rules decreases monotonically. Empirical retention is 0.94-0.96 at $\varepsilon = 0.05$, 0.89-0.90 at $\varepsilon = 0.1$, and 0.76-0.81 at $\varepsilon = 0.2$ across tasks, while the fraction of instances with no guarded rules remains low. These results document high retention and modest sparsification; they do not imply a nominal conformal bound.

In other words, the rules behave as expected, with fewer rules for higher ε , as this translates into a more aggressive removal policy. With $\varepsilon = 0.05$, most rules are retained with only very few instances ending up with no explanations. As ε is increased, the guard removes more rules, reducing retention and increasing the fraction of instances with zero guarded rules. The factual retention rates happen to lie near $1 - \varepsilon$ in these experiments; this is a descriptive observation, not a validity guarantee.

Table 7: Guard retention benchmark for *alternative* explanations across all tasks and regression modes (averaged over datasets, seeds, and calibration sizes). *Standard*: mean number of Calibrated Explanations rules per instance (no guard, at $\varepsilon = 0.10$), *Guarded*: mean number of guarded rules per instance; *Retention*: emitted / (emitted + guard-removed) over alternative bins ($1 - \varepsilon$ is a descriptive comparison only, not a guaranteed floor), *Filtered*: fraction of instances with zero guarded rules.

ε		0.05	0.1	0.2	0.05	0.1	0.2	0.05	0.1	0.2
Task	Standard	<i>Guarded</i>			<i>Retention</i>			<i>Filtered</i>		
Binary	17.0	37.5	34.3	28.1	0.80	0.71	0.55	0.05	0.07	0.14
Multiclass	20.0	70.5	62.8	51.2	0.88	0.77	0.62	0.01	0.02	0.04
Regr. plain	13.5	36.5	32.0	25.3	0.79	0.69	0.55	0.01	0.01	0.04
Regr. p25	12.1	27.9	24.2	19.1	0.76	0.65	0.49	0.01	0.02	0.05
Regr. p50	12.1	27.3	23.6	18.4	0.76	0.64	0.48	0.01	0.02	0.05
Regr. p75	12.3	28.4	24.7	19.4	0.76	0.65	0.49	0.01	0.02	0.05

For alternatives, as shown in Table 7, guarded explanations generally emit many more rules. This is an effect of allowing interval conditions, allowing multiple rules to be emitted for a single numeric feature, representing multiple intervals. Standard alternative rules only emit at most a "lower than" and a "larger than" rule per numeric feature. Again, the results show that ε behaves as expected, with more rules being removed as the conformity requirement becomes stricter. Even under the strictest evaluated conformity requirement ($\varepsilon = 0.2$) in removal decisions, the guarded solution still produces more alternatives than using standard Calibrated Explanations.

For alternatives, where rules represent perturbed instances that may not be provably OOD but may still be non-exchangeable with calibration data, the retention rates are much lower than for factual rules. This is to be expected and highlights the fact that alternative rules must always be considered more as explorative guidance than true alternative options. With the addition of the guard mechanism, this exploration can now be performed under stricter control.

The monotone relationship between ε and filtering is observed for binary classification, multiclass classification, conformal regression, and probabilistic regression. These benchmark results establish behavioral consistency, not plausibility benefit on the real datasets.

6.4. Case study: Guarded Explanations for Automated Real Estate Appraisal

As a practical example of how the guarded explanations contribute to increased intelligibility of machine learning models, we consider a use case from the Norwegian housing market. Financial institutions often rely on *automated valuation models* (AVMs) that estimate the present value of real estate given its location and attributes. State-of-the-art AVMs typically rely on machine learning, which makes XAI tools essential for practical deployment, particularly given the high-stakes decisions that depend on them, such as mortgage origination, credit ratings, and taxation.

We study a dataset of $n = 29\,982$ apartment transactions from Oslo, Norway in 2018–2019. The dataset is provided by the Norwegian financial technology company Eienomsverdi AS, and has been previously studied in the context of model interpretability in Hjort et al. (2024). We train an XGBoost model to predict the transaction price in Norwegian kroner with ten explanatory features: living area (m^2), number of bedrooms, coast distance, geographic coordinates, altitude, build year, site area, neighbourhood density variables, and city district. We reserve 50% of the data for model training, 25% for calibration, and 25% for testing, and wrap the model in a `WrapCalibratedExplainer` to obtain calibrated prediction intervals.

To illustrate the effect of the guard, we explain a single prediction for an apartment in Grünerløkka, one of the central city districts in Oslo. The dwelling has a living area of $49\,m^2$, one bedroom, and was built in 1935. The model prediction is $\hat{y} = 3.504$ million kroner (MNOK) with $[2.693, 4.386]$ MNOK as a calibrated prediction interval. The lower and upper bounds of the predicted interval correspond to the 5th and 95th percentiles, corresponding to a 90% conformal confidence prediction. Figure 1 illustrates a set of factual rules and how much they influence the calibrated prediction for the apartment in question. While many of the feature rules are inconclusive (their interval spans across 0, potentially having either positive or negative impact with a tendency in one or the other direction), the fact that the `LivingArea` is in the range $[41.5, 54.5]\,m^2$ has an unambiguous negative contribution to the price, with the entire interval being clearly below 0. Without the uncertainty intervals, there would not have been any obvious means of knowing whether a factual rule is actually impacting the price or not.

Table 8 shows the conformity audit for two key features. The table shows feature value intervals and the corresponding empirical guard score for the representative perturbation. For living areas, intervals up to roughly $112\,m^2$ receive scores above 0.4, while the largest interval of $> 144.5\,m^2$ is assigned a score of 0.026, indicating weak support relative to the calibration data. Similarly, the largest bedroom bin receives a low score. The audit makes this filtering decision visible to an end user; it does not certify that every value in a retained interval is plausible.

Geography is among the most important explanatory variables in AVMs, and the geographic guard serves as an intuitive visual interpretation of the guard behaviour. In Figure 2 we plot the calibration instances on a map of Oslo and illustrate the conforming region for

```

Calibrated Prediction: 3.504 MNOK
Prediction Interval: [2.693, 4.386]

Factors impacting the calibrated prediction positively:
63.01 < Altitude ≤ 123.9          -- weight ≈ 0.011 [-0.872, 0.821]

Factors impacting the calibrated prediction negatively:
41.5 < LivingArea ≤ 54.5          -- weight ≈ -1.352 [-2.234, -0.541]
262447 < Longitude ≤ 265585      -- weight ≈ -0.223 [-1.105, 0.588]
1934 < BuildYear ≤ 1954          -- weight ≈ -0.187 [-1.069, 0.624]
2510 < CoastDistance ≤ 5680      -- weight ≈ -0.109 [-0.991, 0.702]
3572 < FlatsNearby ≤ 3806        -- weight ≈ -0.066 [-0.948, 0.745]
6650765 < Latitude ≤ 6652153    -- weight ≈ -0.064 [-0.947, 0.746]

```

Figure 1: Factual explanation rules of the predicted value for an apartment in Grünerløkka, one of the central city districts in Oslo.

Table 8: Conformity audit for living area and number of bedrooms for the Grünerløkka test instance (49 m², 1 bedroom). Representatives with guard score $p_{\text{guard}} < \varepsilon$ are excluded.

Feature	Interval	p_{guard}
LivingArea	LivingArea $\leq$ 41.5	0.8568
LivingArea	41.5 < LivingArea $\leq$ 54.5	0.8947
LivingArea	54.5 < LivingArea $\leq$ 70.5	0.7915
LivingArea	70.5 < LivingArea $\leq$ 74.5	0.6998
LivingArea	74.5 < LivingArea $\leq$ 86.5	0.5825
LivingArea	86.5 < LivingArea $\leq$ 112.5	0.4073
LivingArea	112.5 < LivingArea $\leq$ 144.5	0.1925
LivingArea	LivingArea > 144.5	0.0262
Bedrooms	Bedrooms $\leq$ 1.5	0.8946
Bedrooms	1.5 < Bedrooms $\leq$ 2.5	0.8535
Bedrooms	2.5 < Bedrooms $\leq$ 3.5	0.1706
Bedrooms	Bedrooms > 3.5	0.0227

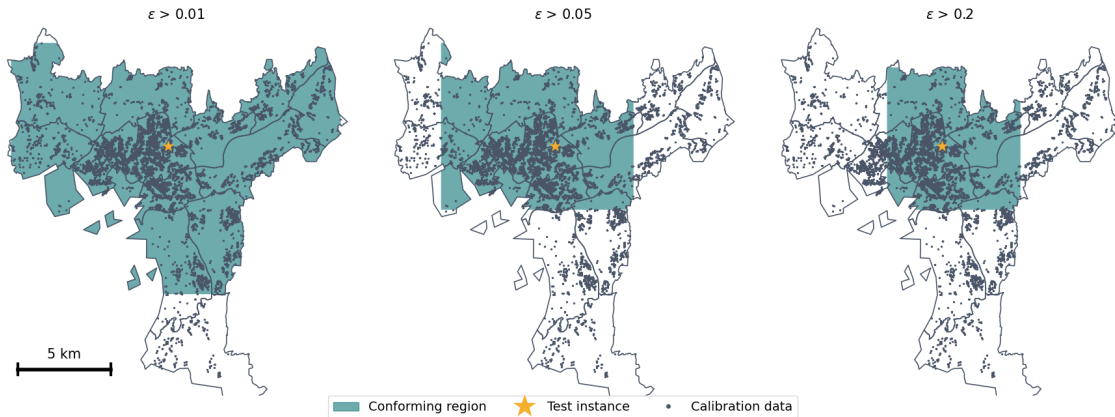
the same test instance at various ε levels. The guard is applied independently on each feature (longitude and latitude), and the visualisation focuses only on non-conformity between the test instance and the calibration instances along one dimension at a time.

At $\varepsilon = 0.01$ the guard is permissive, and the displayed region covers almost the entire city. As the threshold increases, the displayed region contracts around the test instance. Because longitude and latitude are perturbed separately, the map is an illustration of threshold behavior rather than evidence that all shaded coordinate pairs are jointly plausible.

7. Discussion

The rule plausibility evaluation shows that the improvement from standard Calibrated Explanations to guarded Calibrated Explanations is a joint effect of two design changes. Multi-bin discretisation alone already reduces the representative violation rate from ap-

Figure 2: Illustration of how different ε thresholds affect the displayed region. The figure depicts calibration transactions and a test instance marked as a yellow star. The shaded areas combine separate one-feature-at-a-time longitude and latitude checks for $\varepsilon \in \{0.01, 0.05, 0.20\}$; they are not certified joint spatial regions.



proximately 2.5 % (standard Calibrated Explanations) to approximately 0.5 % (multi-bin no-guard), as the ablation in Table 3 confirms. Activating the guard then drives violations from the 0.5 % multi-bin baseline to near zero at $\varepsilon = 0.2$. The guard’s specific contribution is the additional selective filtering of residual non-conforming candidates that multi-bin discretisation alone does not eliminate.

The threshold ε governs the trade-off between plausibility and rule informativeness. In the synthetic constraint setting, $\varepsilon \in [0.1, 0.2]$ achieves near-complete violation suppression while retaining roughly 4.1-4.8 rules per instance on average. This range should not be treated as a general recommendation. The monotonic decline in rule count with increasing ε is a direct consequence of stricter filtering.

Guarded explanations may produce sparse or empty rule sets when an instance lies in a low-density region of the calibration distribution, since few feature-level perturbations will pass the guard. This is the intended behaviour of a selective plausibility filter, not a failure mode. The switch from binary to multi-bin discretisation also changes the rule language independently of the guard, as interval-valued interior bins characterise feature influence regions more precisely than one-sided thresholds, but because the discretisation change is coupled with the guard in our experiments, rule counts in Table 4 are not directly comparable between guarded and standard Calibrated Explanations. Comparison against the multi-bin no-guard baseline in Table 3 controls for this confound.

The guard-score discrimination evaluation confirms that the Fisher-combined guard scores carry distributional signal for moderate-to-extreme shifts across all tested dimensionalities, with mild shifts approaching random ranking at higher dimensions as expected for distance-based KNN scoring. The Fisher-combined scores used in guard-score discrimination evaluation are a diagnostic aggregation of candidate-level guard conformity scores, they are not conformal p-values. The guard-score discrimination evaluation uses controlled shift, not explanation quality or the guard as a deployed anomaly detector.

The scope of the method must be stated precisely. Guarded explanations filter representative perturbations at the per-feature level. Each score nevertheless uses the complete perturbed vector, so the method can respond to feature combinations that are distant from multivariate calibration observations. It does not certify every point in an emitted interval, simultaneous changes to several features, or causal actionability. Constructed perturbations are not exchangeable calibration observations, and the empirical guard score therefore carries no conformal validity guarantee. This caveat is consistent with broader conformal-prediction work showing that distribution shift and departures from exchangeability require explicit care (Tibshirani et al., 2019; Barber et al., 2023). Within these bounds, the guard provides a data-driven mechanism for removing calibration-unsupported probes from emitted factual rules, and the evidence presented here supports that narrow claim.

Evaluation on benchmark datasets extends the behavioral analysis beyond controlled-shift and synthetic settings. By aggregating guard statistics across more than thirty benchmark datasets encompassing binary and multiclass classification and regression tasks, we find that increasing ε consistently reduces retention. In factual mode the average rule count declines only modestly and the fraction of instances with no explanations remains below 0.18 at $\varepsilon = 0.2$. In alternative mode, multi-bin discretisation yields substantially more candidate rules than standard Calibrated Explanations, with the guard acting as a sparsification mechanism. Because these datasets provide no ground-truth labels for perturbation plausibility, the benchmark does not establish that the retained rules are more plausible.

8. Conclusion

We have presented guarded explanations as a selective plausibility mechanism within the Calibrated Explanations framework. The method applies conformal-style rank filtering to candidate perturbation probes and emits only rules built from retained probes, while reusing the Calibrated Explanations prediction backend.

Current results support a narrow but useful claim: guarded filtering can reduce unsupported rules in the evaluated synthetic constraint setting, with explicit sparsity trade-offs and threshold sensitivity. The method should not be interpreted as a guarantee of causal feasibility, semantic validity, joint distribution correctness, or a conformal error rate.

The paper deliberately focuses on two questions only: whether guarded explanations improve rule plausibility under a known constraint, and whether the guard scores themselves show meaningful out-of-distribution discrimination. The synthetic constraint evaluation answers the first question positively at the representative level: guarded explanations reduce emitted-rule violation rates relative to standard Calibrated Explanations, and the ablation shows that the guard provides the final reduction beyond multi-bin discretisation alone. The OOD results support the second question most clearly for moderate-to-extreme shifts, with weaker discrimination for mild shifts in higher dimensions. Future work should evaluate nonlinear, categorical, and multi-feature constraints; explanation faithfulness, stability, and user utility; and sensitivity to k and the distance metric.

The guard is implemented as a Calibrated Explanations compatible extension in the open-source `calibrated-explanations` Python package (https://github.com/Moffran/calibrated_explanations). The package supports binary classification, multiclass classification, and regression; the paper’s headline plausibility result comes from the binary

synthetic constraint setting, complemented by multiclass and regression benchmark evaluations and a real-estate case study.

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